about what constitutes “a measurement” or “an observer”—a measurement is any interaction that causes a quantum system to become entangled with the environment, creating decoherence and a branching into separate worlds, and an observer is any system that brings such an interaction about. Consciousness, in particular, has nothing to do with it. The “observer” could be an earthworm, a microscope, or a rock. There’s not even anything special about macroscopic systems, other than the fact that they can’t help but interact and become entangled with the environment. The price we pay for such powerful and simple unification of quantum dynamics is a large number of separate worlds.

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Everett himself wasn’t familiar with decoherence, so his picture wasn’t quite as robust and complete as the one we’ve painted. But his way of rethinking the measurement problem and offering a unified picture of quantum dynamics was compelling from the start. Even in theoretical physics, people do sometimes get lucky, hitting upon an important idea more because they were in the right place at the right time than because they were particularly brilliant. That’s not the case with Hugh Everett; those who knew him testify uniformly to his incredible intellectual gifts, and it’s clear from his writings that he had a thorough understanding of the implications of his ideas. Were he still alive, he would be perfectly at home in modern discussions of the foundations of quantum mechanics.

What was hard was getting others to appreciate those ideas, and that included his advisor. Wheeler was personally very supportive of Everett, but he was also devoted to his own mentor, Bohr, and was convinced of the basic soundness of the Copenhagen approach. He simultaneously wanted Everett’s ideas to get a wide hearing, and to ensure that they weren’t interpreted as a direct assault on Bohr’s way of thinking about quantum mechanics.

Yet Everett’s theory *was* a direct assault on Bohr’s picture. Everett himself knew it, and enjoyed illustrating the nature of this assault in vivid language. In an early draft of his thesis, Everett used the analogy of an amoeba dividing to illustrate the branching of the wave function: “One can imagine an intelligent amoeba with a good memory. As time progresses the